

■ application-test.properties – Full Explanation

Purpose:

This configuration file is used during **JUnit testing** to prevent connecting to the real MySQL database or sending real emails. Instead, it uses a **temporary H2 in-memory database** and disables email sending.

■ Database Configuration (H2 Memory DB)

```
spring.datasource.url=jdbc:h2:mem:testdb
spring.datasource.driverClassName=org.h2.Driver
spring.datasource.username=sa
spring.datasource.password=
```

■ Creates a temporary H2 database in memory. No password needed. Perfect for isolated test runs.

■■ Hibernate Settings

```
spring.jpa.hibernate.ddl-auto=create-drop
spring.jpa.show-sql=false
```

■ Automatically creates tables when tests start and drops them after completion. SQL logs are hidden for cleaner output.

■ Disable Email Sending

```
spring.mail.host=localhost
spring.mail.port=2525
spring.mail.username=test
spring.mail.password=test
spring.mail.properties.mail.smtp.auth=false
spring.mail.properties.mail.smtp.starttls.enable=false
```

■ Prevents sending real emails during automated tests. Uses dummy values.

■ Spring Boot Flexibility

```
spring.main.allow-bean-definition-override=true
```

■ Allows Spring beans with the same name to be overridden, useful in testing environments.

■ Summary Table

Section	Purpose	Effect
H2 Database	Temporary in-memory DB	No real database used
Hibernate	Auto table create/drop	Ensures clean test environment
Mail	Disabled	Prevents real email sending
Bean Override	Allowed	Avoids bean conflicts during tests

■ When you run **mvn test**, Spring Boot automatically uses this configuration, so your real MySQL DB and email system remain untouched.